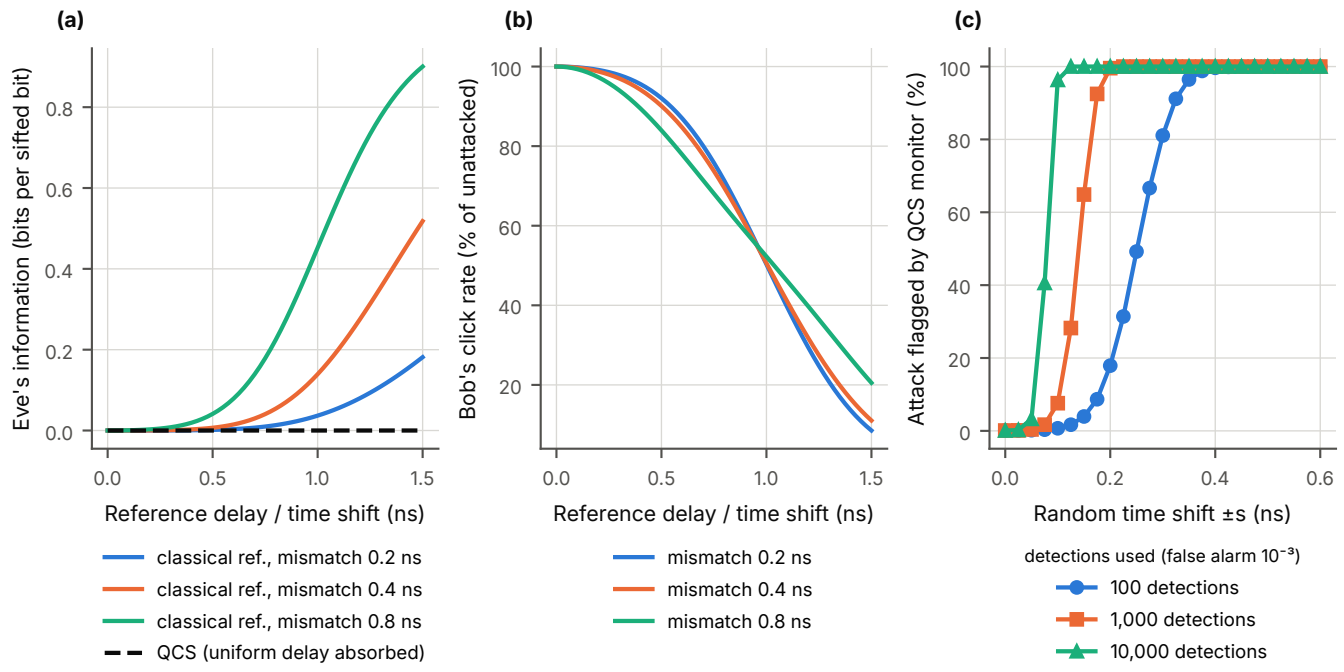
(a)
Eve's information (bits per sifted bit)
0.0
0.2
0.4
0.6
0.8
0.0
0.5
1.0
1.5
Reference delay / time shift (ns)
classical ref., mismatch 0.2 ns
classical ref., mismatch 0.4 ns
classical ref., mismatch 0.8 ns
QCS (uniform delay absorbed)
(b)
Bob's click rate (% of unattacked)
100
80
60
40
20
0.0
0.5
1.0
1.5
Reference delay / time shift (ns)
mismatch 0.2 ns
mismatch 0.4 ns
mismatch 0.8 ns
(c)
Attack flagged by QCS monitor (%)
100
80
60
40
20
0
0.0
0.2
0.4
0.6
Random time shift ±s (ns)
detections used (false alarm 10⁻³)
100 detections
1,000 detections
10,000 detections